

MANAUS PONTA PELADA, Brazil

WMO: 823320

Lat: 3.146S

Lon: 59.986W

Elev: 81

StdP: 100.35

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.3	22.9	19.1	14.0	29.2	20.0	14.9	28.7	6.7	26.7	5.9	27.6	2.1	60	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	7.9	35.0	25.8	34.2	25.8	33.7	25.8	27.2	31.6	26.9	31.3	26.6	31.0	3.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	29.3	25.9	21.4	29.1	25.3	20.7	28.5	86.7	31.8	85.0	31.7	83.7	31.4	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.8	4.2	DB	20.5	36.7	1.0	1.3	19.8	37.7	19.2	38.4	18.6	39.1	17.8	40.1	(4)
(5)			WB	19.7	28.2	1.3	1.1	18.8	29.0	18.1	29.6	17.4	30.2	16.5	31.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.1	27.0	27.0	27.2	27.3	27.5	27.9	28.7	29.1	28.9	28.4	27.6	(6)	
	DBStd	1.56	1.37	1.47	1.34	1.23	1.14	1.21	1.23	1.28	1.50	1.60	1.51	1.47	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6503	531	476	528	517	536	525	554	581	572	587	552	544	(10)	
	CDD18.3	3461	272	243	270	267	277	275	296	322	322	329	302	286	(11)	
	CDH23.3	35599	2421	2117	2364	2424	2640	2795	3181	3781	3892	3845	3348	2792	(12)	
	CDH26.7	12777	704	602	646	672	745	899	1153	1609	1768	1706	1359	914	(13)	
(14) Wind	WSAvg	1.8	1.8	1.9	1.8	1.6	1.5	1.6	1.7	1.8	1.9	1.9	1.8	1.8	(14)	
(15) Precipitation	PrecAvg	2288	284	285	326	319	243	116	76	56	73	115	182	238	(15)	
	PrecMax	3156	570	619	633	535	555	228	205	141	243	253	337	519	(16)	
	PrecMin	1764	33	108	134	136	66	10	3	2	0	11	13	90	(17)	
	PrecStd	292	90	110	108	98	98	47	47	40	52	62	86	101	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	33.5	33.2	33.5	32.8	33.0	33.9	34.9	36.1	35.9	35.1	34.1	(19)	
		MCWB	25.9	26.3	26.5	26.8	26.5	26.0	25.6	26.1	25.2	25.4	25.7	25.8	(20)	
	2%	DB	32.1	32.1	32.0	32.0	31.9	32.0	32.8	34.0	35.0	34.9	34.1	33.0	(21)	
		MCWB	25.7	25.7	26.0	26.1	26.1	25.7	25.4	25.6	25.6	25.8	25.8	25.7	(22)	
	5%	DB	31.1	31.1	31.0	31.0	31.0	31.2	32.0	33.1	34.0	34.0	33.1	31.9	(23)	
		MCWB	25.6	25.6	25.9	26.0	26.0	25.7	25.3	25.5	25.7	25.8	25.9	25.8	(24)	
	10%	DB	30.1	30.0	30.0	30.1	30.1	30.8	31.2	32.4	33.1	33.0	32.1	30.9	(25)	
		MCWB	25.5	25.4	25.6	25.8	25.8	25.6	25.3	25.4	25.6	25.7	25.8	25.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.3	27.3	27.6	27.2	26.9	26.7	27.1	27.2	27.4	27.2	27.2	(27)	
		MCDB	31.0	31.3	31.4	31.4	30.9	30.9	31.2	32.2	32.3	32.4	32.0	31.2	(28)	
	2%	WB	26.4	26.6	26.6	26.7	26.6	26.4	26.2	26.6	26.7	26.9	26.8	26.6	(29)	
		MCDB	30.1	30.2	30.3	30.4	30.1	30.3	30.7	31.7	31.9	32.1	31.5	30.5	(30)	
	5%	WB	26.1	26.1	26.1	26.2	26.2	26.1	26.0	26.2	26.4	26.5	26.5	26.2	(31)	
		MCDB	29.6	29.5	29.6	29.5	29.5	29.9	30.4	31.2	31.5	31.6	31.0	29.9	(32)	
	10%	WB	25.7	25.7	25.8	25.9	26.0	25.9	25.7	25.9	26.1	26.2	26.2	25.9	(33)	
		MCDB	29.0	28.8	28.9	29.1	29.2	29.5	29.9	30.8	31.1	31.1	30.5	29.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	6.0	6.0	5.9	6.3	6.9	7.6	7.9	7.8	7.2	6.4	(35)	
		MCDBR	7.4	7.4	7.4	7.3	6.9	7.0	7.6	8.2	8.8	8.8	8.3	7.8	(36)	
	5% WB	MCWBR	2.8	2.8	3.0	3.0	2.7	2.7	2.8	2.9	3.1	2.9	2.9	2.9	(37)	
		MCWBR	6.6	6.6	6.8	6.5	6.3	6.6	7.1	7.8	8.1	8.1	7.6	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.399	0.393	0.393	0.385	0.387	0.390	0.403	0.488	0.529	0.508	0.494	0.452	(40)	
		taud	2.520	2.546	2.553	2.585	2.568	2.560	2.498	2.221	2.089	2.170	2.207	2.348	(41)	
	Ebn at Noon		935	940	930	914	887	871	864	811	800	828	841	881	(42)	
		Edh at Noon	111	109	107	100	98	97	104	143	168	157	151	131	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.30	4.27	4.28	4.21	4.15	4.60	4.94	5.30	5.30	5.06	4.67	4.20	(44)	
	RadStd		0.35	0.33	0.32	0.16	0.26	0.28	0.24	0.30	0.40	0.27	0.31	0.35	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	-0.22	-0.11	-1.63	-0.52	-1.15	-1.26	N/A	N/A	-153	-153

Nomenclature: See separate page